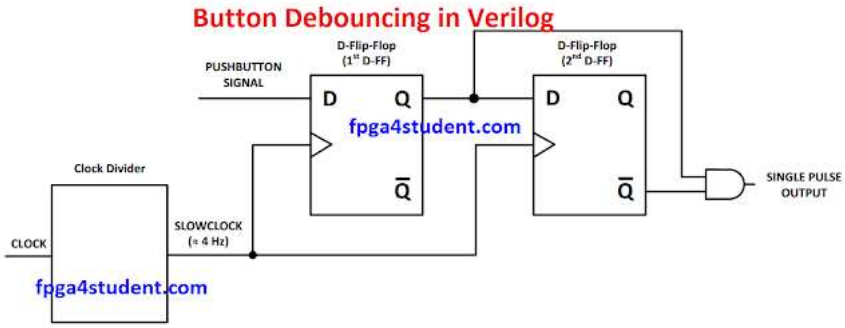
Verilog code for debouncing buttons on FPGA

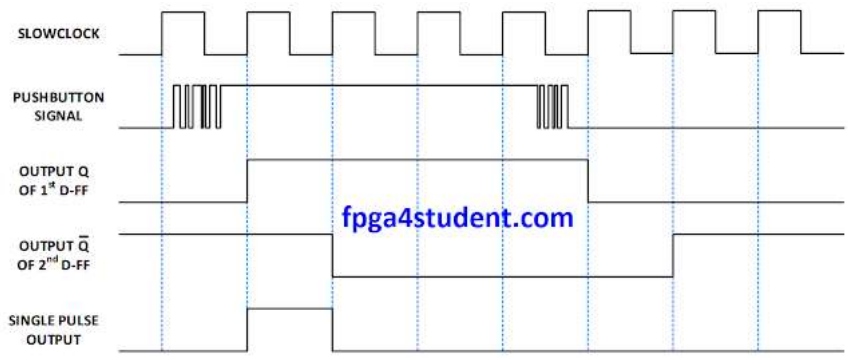
This post is to present a simple debouncing Verilog code for buttons on FPGA.

Mechanical switches/ buttons cause an unpredictable bounce in the signal when toggled. There are various ways to implement debouncing circuits for buttons on FPGA. In this project, a simple debouncing circuit is implemented in Verilog to generate only a single pulse when pressing a button on FPGA.



Debouncing Circuit for buttons on FPGA

As shown in the figure below, when a button on FPGA is pressed and released, there are many unexpected up-and-down bounces in the push-button signal. The debouncing circuit only generates a single pulse with a period of the slow clock without bouncing as we expected.



Expected waveform from the debouncing circuit

Verilog code for button debouncing on FPGA (extra FF to avoid metastability for the 1st FF):

```
//fpga4student.com
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// Verilog code for button debouncing on FPGA
// debouncing module
module debounce(input pb_1,clk,output pb_out);
wire slow_clk;
wire Q1,Q2,Q2_bar,Q0;
clock_div u1(clk,slow_clk);

my_dff d0(slow_clk, pb_1,Q0 );


my_dff d1(slow_clk, Q0,Q1 );
my_dff d2(slow_clk, Q1,Q2 );
assign Q2_bar = ~Q2;
assign pb_out = Q1 & Q2_bar;
endmodule
// Slow clock for debouncing
module clock_div(input Clk_100M, output reg slow_clk

    );
    reg [26:0]counter=0;
```
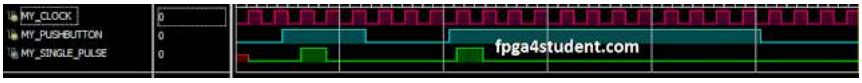

```
    always @(posedge Clk_100M)
    begin
        counter <= (counter>=249999)?0:counter+1;
        slow_clk <= (counter < 125000)?1'b0:1'b1;
    end
endmodule
// D-flip-flop for debouncing module
module my_dff(input DFF_CLOCK, D, output reg Q);

    always @ (posedge DFF_CLOCK) begin
        Q <= D;
    end

endmodule
```

Simulation waveform for button debouncing (2FFs):



As shown in the waveform, only a single pulse is generated when a button is pressed and released as expected.

It is noted that this code is about to create another clock in your design, so the FPGA tools required to take care of an extra internally generated clock during clock tree synthesis, which might cause FPGA timing issues as it is not generated by dedicated FPGA clock generators (PLL/DCM/etc). In addition, you also have to take care of the multi-clock domain issues while designing such as interfacing signals between different clock domains (synchronizer needed, etc). Instead of creating another slow clock in the design, we can generate the clock enable signal to drive the debouncing flip-flops.

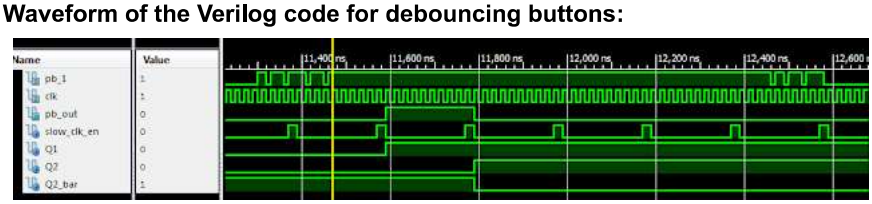
The better Verilog code for debouncing buttons on FPGA without creating another clock domain:

```
//fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// Verilog code for button debouncing on FPGA
// debouncing module without creating another clock domain
// by using clock enable signal
module debounce_better_version(input pb_1,clk,output pb_out);
wire slow_clk_en;
wire Q1,Q2,Q2_bar,Q0;
clock_enable u1(clk,slow_clk_en);

my_dff_en d0(clk,slow_clk_en,pb_1,Q0);


my_dff_en d1(clk,slow_clk_en,Q0,Q1);
my_dff_en d2(clk,slow_clk_en,Q1,Q2);
assign Q2_bar = ~Q2;
assign pb_out = Q1 & Q2_bar;
endmodule
// Slow clock enable for debouncing button
module clock_enable(input Clk_100M,output slow_clk_en);
    reg [26:0]counter=0;
    always @(posedge Clk_100M)
    begin
       counter <= (counter>=249999)?0:counter+1;
    end
    assign slow_clk_en = (counter == 249999)?1'b1:1'b0;
endmodule
// D-flip-flop with clock enable signal for debouncing module
module my_dff_en(input DFF_CLOCK, clock_enable,D, output reg Q=0);
    always @ (posedge DFF_CLOCK) begin
  if(clock_enable==1)
            Q <= D;
    end
endmodule
```

Testbench Verilog code for debouncing buttons:

```
`timescale 1ns / 1ps
// testbench verilog code for debouncing button without creating another clock
module tb_button;
 // Inputs
 reg pb_1;
 reg clk;
 // Outputs
 wire pb_out;
 // Instantiate the debouncing Verilog code
 debounce_better_version uut (
  .pb_1(pb_1),
  .clk(clk),
  .pb_out(pb_out)
 );
 initial begin
  clk = 0;
  forever #10 clk = ~clk;
 end
 initial begin
  pb_1 = 0;
  #10;
  pb_1=1;
  #20;
  pb_1 = 0;
  #10;
  pb_1=1;
  #30;
  pb_1 = 0;
  #10;
  pb_1=1;
  #40;
  pb_1 = 0;
  #10;
  pb_1=1;
  #30;
  pb_1 = 0;
  #10;
  pb_1=1;
  #1000;
  pb_1 = 0;
  #10;
  pb_1=1;
  #20;
  pb_1 = 0;
  #10;
  pb_1=1;
  #30;
  pb_1 = 0;
  #10;
  pb_1=1;
  #40;
  pb_1 = 0;
 end

endmodule
```

Simulation Waveform of the Verilog code for debouncing buttons:



As shown in the simulation waveform, only a pulse with a period of the slow clock enable signal is generated when a button is pressed, held enough, and released as expected. Note that the divisor of 9 is used for faster simulation. If you want a single pulse with a period of the clock input to be generated, simply modify the output assignment "*assign pb_out = Q1 & Q2_bar & slow_clk_en;*".

VHDL code for debouncing buttons on FPGA

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